

MCS603

Python Programming

230 Multiple Choice Questions

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Total Questions	230 MCQs (11 Sections — Python + Streamlit)

Sections covered:

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2	Data Types & Structures	25	Q31 – Q55
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SECTION 1 — Python Fundamentals

- Q1. What does the # symbol do in Python?
- A) Declares a variable
 - B) Multiplication operator
 - C) Marks the beginning of a comment
 - D) Modulo operator
- Q2. Which of the following is a valid Python variable name?
- A) 2myvar
 - B) my-var
 - C) my_var
 - D) my var
- Q3. What is the output of: `print(10 // 3)`?
- A) 3.33
 - B) 3
 - C) 4
 - D) 1
- Q4. Which keyword is used to define a function in Python?
- A) function
 - B) define
 - C) def
 - D) func
- Q5. What is the correct file extension for Python files?
- A) .pt
 - B) .py
 - C) .pyt
 - D) .python
- Q6. What does `print(10 % 3)` output?
- A) 3
 - B) 0
 - C) 1
 - D) 3.33
- Q7. Which of the following is NOT a Python keyword?
- A) for
 - B) while
 - C) loop
 - D) if
- Q8. What is the output of: `print(2 ** 8)`?
- A) 16

- B) 64
- C) 256
- D) 512

Q9. What does `len("hello")` return?

- A) 4
- B) 5
- C) 6
- D) 'hello'

Q10. Which operator is used for string concatenation in Python?

- A) &
- B) .
- C) +
- D) *

Q11. What is the output of: `print("abc" * 2)`?

- A) abc2
- B) abcab
- C) 6
- D) Error

Q12. What does the `type()` function return?

- A) The size of a variable
- B) The data type of a variable
- C) The value of a variable
- D) The name of a variable

Q13. Which statement exits a loop prematurely?

- A) `exit`
- B) `stop`
- C) `break`
- D) `quit`

Q14. What does `bool(0)` evaluate to?

- A) True
- B) False
- C) 0
- D) None

Q15. What does `int("42")` return?

- A) "42"
- B) 42
- C) 42.0
- D) Error

Q16. Which of the following creates an empty list?

- A) `list = {}`
- B) `list = []`

- C) list = ()
- D) list = <>

Q17. What is the result of `3 == 3.0` in Python?

- A) False
- B) True
- C) Error
- D) None

Q18. Which Python version introduced f-strings?

- A) Python 2.7
- B) Python 3.0
- C) Python 3.6
- D) Python 3.9

Q19. What does the `pass` statement do?

- A) Exits the function
- B) Acts as a no-op placeholder
- C) Passes a value to the next line
- D) Skips the current iteration

Q20. Which method converts a string to lowercase?

- A) `str.lower()`
- B) `str.toLower()`
- C) `str.lowercase()`
- D) `str.small()`

Q21. What is the output of `print(not True)`?

- A) True
- B) 1
- C) False
- D) Not True

Q22. What does `range(5)` generate?

- A) [1,2,3,4,5]
- B) [0,1,2,3,4]
- C) (0,1,2,3,4,5)
- D) {0,1,2,3,4}

Q23. Which of the following is a mutable data type?

- A) tuple
- B) string
- C) list
- D) int

Q24. What does the `input()` function always return?

- A) An integer
- B) A float
- C) A string

D) A boolean

Q25. What does `abs(-7)` return?

- A) -7
- B) 7
- C) 0
- D) Error

Q26. What is the output of `print(10 > 5 and 3 < 2)`?

- A) True
- B) False
- C) Error
- D) None

Q27. Which built-in function returns the smaller of two values?

- A) `small()`
- B) `minimum()`
- C) `min()`
- D) `lower()`

Q28. What does `round(3.567, 2)` return?

- A) 3.5
- B) 3.56
- C) 3.57
- D) 3.6

Q29. What is the output of `print(type({}))`?

- A) `<class 'list'>`
- B) `<class 'set'>`
- C) `<class 'dict'>`
- D) `<class 'tuple'>`

Q30. Which symbol represents the exponentiation operator in Python?

- A) `^`
- B) `exp`
- C) `**`
- D) `^^`

SECTION 2 — Data Types & Structures

- Q31. Which of the following creates a tuple?
- A) `t = [1,2,3]`
 - B) `t = {1,2,3}`
 - C) `t = (1,2,3)`
 - D) `t = <1,2,3>`
- Q32. What is the index of the first element in a Python list?
- A) 1
 - B) 0
 - C) -1
 - D) None
- Q33. What does `list.append(x)` do?
- A) Inserts `x` at position 0
 - B) Removes `x` from the list
 - C) Adds `x` to the end of the list
 - D) Replaces the first element with `x`
- Q34. What is a Python dictionary?
- A) An ordered sequence
 - B) An unordered collection of key-value pairs
 - C) An immutable list
 - D) A set of unique integers
- Q35. How do you access the value of key `name` in dict `d`?
- A) `d.name`
 - B) `d["name"]`
 - C) `d->name`
 - D) `d.get_value("name")`
- Q36. What does `set({1, 2, 2, 3})` produce?
- A) `{1,2,2,3}`
 - B) `{1,2,3}`
 - C) `[1,2,3]`
 - D) `(1,2,3)`
- Q37. Which method removes the last item from a list?
- A) `list.remove()`
 - B) `list.delete()`
 - C) `list.pop()`
 - D) `list.drop()`
- Q38. What is the output of `"hello"[1]`?
- A) `h`

- B) e
- C) hel
- D) ello

Q39. What does `dict.keys()` return?

- A) All values
- B) All keys
- C) All key-value pairs
- D) The number of keys

Q40. What is the correct way to create an empty set?

- A) `s = []`
- B) `s = {}`
- C) `s = set()`
- D) `s = ()`

Q41. What is the result of `[1,2,3] + [4,5,6]`?

- A) `[5,7,9]`
- B) `[1,2,3,4,5,6]`
- C) Error
- D) `[10,5,9]`

Q42. Which method returns all values from a dictionary?

- A) `dict.items()`
- B) `dict.keys()`
- C) `dict.values()`
- D) `dict.all()`

Q43. What does `tuple.count(x)` do?

- A) Returns index of x
- B) Returns times x appears in the tuple
- C) Counts all elements
- D) Returns True if x is in the tuple

Q44. What is a list comprehension?

- A) A way to sort a list
- B) A concise way to create lists using a for loop inline
- C) A method to compress a list
- D) A way to merge two lists

Q45. What does `sorted([3,1,2])` return?

- A) `[3,2,1]`
- B) `[1,2,3]`
- C) `{1,2,3}`
- D) `(1,2,3)`

Q46. How do you check if a key exists in a dictionary `d`?

- A) `key in d`
- B) `d.hasKey(key)`

- C) `d.exists(key)`
- D) `key.in(d)`

Q47. What is the output of `len({"a":1, "b":2})`?

- A) 1
- B) 2
- C) 4
- D) 0

Q48. Which of the following is immutable?

- A) list
- B) dict
- C) set
- D) tuple

Q49. What does `list.insert(i, x)` do?

- A) Replaces element at index `i` with `x`
- B) Inserts `x` before index `i`
- C) Adds `x` at the end
- D) Removes at `i` and inserts `x`

Q50. What is the output of `[0] * 3`?

- A) `[0,0,0]`
- B) `[3]`
- C) `[0,3]`
- D) Error

Q51. Which method sorts a list in-place?

- A) `sorted()`
- B) `list.sort()`
- C) `list.order()`
- D) `list.arrange()`

Q52. What does `"hello".upper()` return?

- A) `"HELLO"`
- B) `"Hello"`
- C) `"hello"`
- D) `"hELLO"`

Q53. What is the output of `(1,2,3)[1:]`?

- A) `(1,2)`
- B) `(2,3)`
- C) `(1,2,3)`
- D) `[2,3]`

Q54. Which method removes a key from a dictionary?

- A) `dict.remove(key)`
- B) `dict.delete(key)`
- C) `del dict[key]`

D) dict.discard(key)

Q55. What is a frozenset?

- A) A list that cannot be modified
- B) An immutable version of a set
- C) A sorted set
- D) A dictionary with frozen keys

SECTION 3 — Control Flow

Q56. Which keyword handles a chain of conditions after `if`?

- A) `else`
- B) `elif`
- C) `elif`
- D) `elseif`

Q57. What does a `while` loop do?

- A) Executes code a fixed number of times
- B) Executes code while a condition is `True`
- C) Iterates over a sequence
- D) Runs code once

Q58. What does the `continue` statement do?

- A) Exits the loop
- B) Skips rest of current iteration and continues to next
- C) Pauses the loop
- D) Restarts the loop

Q59. What does `range(1, 10, 2)` produce?

- A) `[1,3,5,7,9]`
- B) `[1,2,3,4,5,6,7,8,9]`
- C) `[2,4,6,8]`
- D) `[1,10,2]`

Q60. What is the purpose of the `else` clause in a `for` loop?

- A) Runs if loop was interrupted by `break`
- B) Runs after loop completes without `break`
- C) Handles exceptions
- D) Not valid in Python

Q61. What does the `break` statement do?

- A) Skips the current iteration
- B) Exits the innermost loop
- C) Exits the entire program
- D) Raises an exception

Q62. What is a nested loop?

- A) A loop inside a function
- B) A loop inside another loop
- C) A loop with multiple conditions
- D) A loop that runs infinitely

Q63. What is the ternary expression syntax in Python?

- A) `x if condition else y`

- B) condition ? x : y
- C) if condition then x else y
- D) condition : x | y

Q64. What does `while True:` create?

- A) A loop that runs once
- B) An infinite loop
- C) A loop that never runs
- D) A conditional loop

Q65. Which of the following is a valid Python for loop?

- A) `for (i=0;i<5;i++):`
- B) `for i in range(5):`
- C) `for i = 0 to 5:`
- D) `foreach i in range(5):`

Q66. How many times does `for i in range(0,10,3)` execute?

- A) 3
- B) 4
- C) 10
- D) 5

Q67. What does `enumerate(["a","b","c"])` yield?

- A) `["a","b","c"]`
- B) `(0,"a"),(1,"b"),(2,"c")`
- C) `["a0","b1","c2"]`
- D) `{0:"a",1:"b",2:"c"}`

Q68. What is the output of `for i in range(3): print(i)`?

- A) 1 2 3
- B) 0 1 2
- C) 0 1 2 3
- D) 1 2

Q69. Which statement causes an infinite loop?

- A) `while False:`
- B) `for i in range(0):`
- C) `while True: break`
- D) `while 1 == 1:`

Q70. What does `pass` do in an if block?

- A) Skips the entire if block
- B) Acts as a no-op placeholder
- C) Exits the program
- D) Continues from the next if

Q71. What is the output of the expression: `5 if 10>5 else 0`?

- A) 0
- B) 10

- C) 5
- D) Error

Q72. What does `zip([1,2],[3,4])` return?

- A) `[4,6]`
- B) `[(1,3),(2,4)]`
- C) `[1,2,3,4]`
- D) `{1:3,2:4}`

Q73. Which loop is guaranteed to execute at least once?

- A) for loop
- B) while loop
- C) do-while (not in Python natively)
- D) None of the above

Q74. What is the output of `list(range(2,8,2))`?

- A) `[2,4,6]`
- B) `[2,4,6,8]`
- C) `[2,3,4,5,6,7,8]`
- D) `[0,2,4,6,8]`

Q75. What does `any([False,False,True])` return?

- A) False
- B) True
- C) None
- D) Error

SECTION 4 — Functions & Lambda

Q76. What does a function return if there is no `return` statement?

- A) 0
- B) False
- C) None
- D) Error

Q77. What is a default parameter?

- A) A parameter with a predefined value
- B) The first parameter
- C) A required parameter
- D) A parameter that cannot change

Q78. What does `*args` allow in a function?

- A) Only keyword arguments
- B) A variable number of positional arguments
- C) A single argument
- D) Named arguments only

Q79. What does `**kwargs` allow in a function?

- A) Variable number of positional arguments
- B) A variable number of keyword arguments
- C) Keyword-only arguments with defaults
- D) All types of arguments

Q80. What is a lambda function?

- A) A function with multiple lines
- B) An anonymous single-expression function
- C) A function that calls itself
- D) A function with no parameters

Q81. What is the output of `(lambda x: x*2)(5)`?

- A) 5
- B) 10
- C) 25
- D) 52

Q82. What is a recursive function?

- A) A function that calls another function
- B) A function that calls itself
- C) A function with many parameters
- D) A function that never returns

Q83. What is the scope of a variable defined inside a function?

- A) Global

- B) Local
- C) Module
- D) Built-in

Q84. What does the `global` keyword do inside a function?

- A) Makes variable local only
- B) Declares a variable as global
- C) Imports a global module
- D) Deletes a global variable

Q85. What does the `map()` function do?

- A) Creates a dictionary
- B) Applies a function to every item in an iterable
- C) Filters items from a list
- D) Reduces a list to a single value

Q86. What does `filter(func, iterable)` return?

- A) All items from iterable
- B) Items where `func` returns `True`
- C) Items where `func` returns `False`
- D) The result of applying `func` to each item

Q87. What does `@` before a function definition denote?

- A) A comment
- B) A decorator
- C) An import statement
- D) A class method marker

Q88. What is a closure in Python?

- A) A way to close a file
- B) A function retaining access to enclosing scope variables
- C) A function with no return value
- D) An exception handler

Q89. What does `all([True, True, False])` return?

- A) `True`
- B) `False`
- C) `None`
- D) Error

Q90. What does `sorted(iterable, reverse=True)` do?

- A) Sorts ascending
- B) Sorts descending
- C) Shuffles the iterable
- D) Reverses without sorting

Q91. What is the output of `max(3, 7, 2)`?

- A) 3
- B) 2

- C) 7
- D) 12

Q92. What does `reduce(lambda x,y: x+y, [1,2,3,4])` return?

- A) [1,3,6,10]
- B) 10
- C) [10]
- D) 24

Q93. Which built-in function converts a value to a string?

- A) `toString()`
- B) `string()`
- C) `str()`
- D) `text()`

Q94. What does `isinstance(42, int)` return?

- A) False
- B) True
- C) 42
- D) int

Q95. What does `sum([1,2,3,4])` return?

- A) 10
- B) [1,2,3,4]
- C) 4
- D) Error

SECTION 5 — Modules & Packages

- Q96. What is a Python module?
- A) A single Python function
 - B) A file containing Python code that can be imported
 - C) A compiled Python program
 - D) A Python class
- Q97. What is a Python package?
- A) A compressed archive
 - B) A directory containing `__init__.py` and modules
 - C) A single module
 - D) An installed library
- Q98. What does `import math` do?
- A) Imports a specific function
 - B) Imports the entire math module
 - C) Installs the math library
 - D) Compiles math module
- Q99. What does `from math import sqrt` do?
- A) Imports the entire math module
 - B) Imports only `sqrt` from `math`
 - C) Creates a new `sqrt` function
 - D) Renames `math` to `sqrt`
- Q100. What does `__init__.py` do in a package directory?
- A) Initialises the Python interpreter
 - B) Marks the directory as a Python package
 - C) Lists installed packages
 - D) Configures the Python path
- Q101. What is the purpose of `pip`?
- A) A Python debugger
 - B) A Python package installer
 - C) A Python linter
 - D) A Python compiler
- Q102. How do you install a package with `pip`?
- A) `pip get pandas`
 - B) `pip install pandas`
 - C) `pip add pandas`
 - D) `pip download pandas`
- Q103. What command lists all installed packages?
- A) `pip list`

- B) pip show
- C) pip packages
- D) pip all

Q104. What does `import os` give access to?

- A) Object-oriented tools
- B) Operating system interface functions
- C) Online services
- D) Output stream functions

Q105. What does `from . import module` mean?

- A) Import from built-in library
- B) Import from the current package (relative import)
- C) Import from parent directory
- D) Import from global scope

Q106. What does `__name__ == "__main__"` check?

- A) Whether the module is empty
- B) Whether the script is run directly (not imported)
- C) Whether the module has a main function
- D) Whether the module is installed

Q107. Which module provides regular expressions?

- A) regex
- B) re
- C) regexp
- D) pattern

Q108. How do you add a directory to `sys.path`?

- A) `sys.path.import(dir)`
- B) `sys.path.insert(0, dir)`
- C) `sys.add(dir)`
- D) `path.append(sys, dir)`

Q109. What does `__all__` in a module control?

- A) All class attributes
- B) Names exported with `'from module import *'`
- C) All global variables
- D) All import statements

Q110. What is a virtual environment?

- A) A simulated computer
- B) An isolated Python environment with its own packages
- C) A cloud-based runtime
- D) A Python debugger

Q111. Which command creates a virtual environment named `venv`?

- A) `python -m venv create venv`
- B) `python -m venv venv`

- C) `venv create python`
- D) `pip create venv`

Q112. What file lists a project's dependencies for pip?

- A) `dependencies.txt`
- B) `packages.txt`
- C) `requirements.txt`
- D) `install.txt`

Q113. What does `from ..module import func` do?

- A) Imports from the current package
- B) Imports from the parent package
- C) Imports from a sub-package
- D) Imports from the built-in library

Q114. What does `dir(module)` return?

- A) The file path of the module
- B) A list of names defined in the module
- C) The documentation
- D) The version

Q115. Which of the following is NOT part of Python's standard library?

- A) `os`
- B) `sys`
- C) `numpy`
- D) `math`

Q116. What does `import numpy as np` do?

- A) Imports numpy and renames it to `np`
- B) Installs numpy with alias `np`
- C) Imports function '`np`' from numpy
- D) Creates a shortcut for installation

Q117. How do you view documentation for a module's function?

- A) `doc(module.function)`
- B) `help(module.function)`
- C) `info(module.function)`
- D) `describe(module.function)`

Q118. What does `sys.argv` contain?

- A) System configuration
- B) Command-line arguments passed to the script
- C) Available system modules
- D) Python version info

Q119. What does `importlib.reload(module)` do?

- A) Installs a fresh copy of the module
- B) Reloads a previously imported module
- C) Uninstalls the module

D) Copies the module

Q120. Which module provides namedtuple?

- A) tuple
- B) namedtuple
- C) collections
- D) typing

SECTION 6 — File I/O

Q121. What does `open("file.txt", "r")` do?

- A) Creates a new file
- B) Opens a file for reading
- C) Opens a file for writing
- D) Opens a file for appending

Q122. What mode opens a file for writing (overwrites existing)?

- A) "r"
- B) "a"
- C) "w"
- D) "x"

Q123. What mode appends data to an existing file?

- A) "r"
- B) "w"
- C) "a"
- D) "rw"

Q124. What does `file.read()` do?

- A) Reads one character
- B) Reads one line
- C) Reads the entire file as a string
- D) Reads specified bytes

Q125. What does `file.readlines()` do?

- A) Reads the entire file
- B) Reads one line at a time
- C) Reads all lines as a list
- D) Reads from the end of file

Q126. What is the recommended way to open a file?

- A) `file = open("f.txt")`
- B) `with open("f.txt") as f:`
- C) `f = File("f.txt")`
- D) `f.open("f.txt")`

Q127. What does the `with` statement ensure for file operations?

- A) The file opens faster
- B) The file is automatically closed
- C) The file is read-only
- D) The file is cached

Q128. What does `file.readlines()` return?

- A) A string of the whole file

- B) A list of all lines
- C) An integer count of lines
- D) A generator object

Q129. Which module provides functions for working with file paths?

- A) filepath
- B) os.path
- C) sys.path
- D) filePath

Q130. What does `os.path.exists("file.txt")` return?

- A) The size of the file
- B) True if path exists False otherwise
- C) The content of the file
- D) The creation date

Q131. Which module is used for working with CSV files?

- A) filetools
- B) csv
- C) table
- D) spreadsheet

Q132. What does `json.dumps(data)` do?

- A) Loads JSON from a file
- B) Converts Python object to a JSON string
- C) Saves to a JSON file
- D) Parses a JSON string

Q133. What does `json.loads(string)` do?

- A) Saves a string to JSON file
- B) Parses a JSON string into a Python object
- C) Converts Python to JSON
- D) Loads all JSON files

Q134. What does `os.getcwd()` return?

- A) List of files
- B) The current working directory path
- C) The OS name
- D) The home directory

Q135. What exception does `open("nonexistent.txt")` raise?

- A) FileNotFoundError
- B) IOError
- C) FileNotNotFoundError
- D) ReadError

SECTION 7 — Exception Handling

Q136. What keyword is used to catch exceptions?

- A) catch
- B) except
- C) handle
- D) error

Q137. What does the finally block do?

- A) Runs only if no exception
- B) Runs only if exception occurs
- C) Always runs regardless of exception
- D) Catches all exceptions

Q138. How do you raise a custom exception?

- A) throw Exception("msg")
- B) raise Exception("msg")
- C) error Exception("msg")
- D) trigger Exception("msg")

Q139. What is ZeroDivisionError?

- A) Undefined variable
- B) Dividing by zero
- C) Index out of range
- D) Key missing from dict

Q140. What exception is raised when a list index is out of range?

- A) KeyError
- B) ValueError
- C) IndexError
- D) TypeError

Q141. What exception is raised when a dict key is not found?

- A) IndexError
- B) KeyError
- C) ValueError
- D) NotFoundException

Q142. What does the else clause in a try block do?

- A) Runs if an exception is raised
- B) Runs if no exception is raised
- C) Always runs after try
- D) Handles a specific exception

Q143. What exception is raised when a variable is not defined?

- A) TypeError

- B) AttributeError
- C) NameError
- D) ValueError

Q144. What does `except Exception as e` do?

- A) Raises exception `e`
- B) Catches all exceptions and binds to `e`
- C) Catches only `Exception` type
- D) Prints exception `e`

Q145. What exception is raised by `int("abc")`?

- A) `TypeError`
- B) `ValueError`
- C) `CastError`
- D) `ConversionError`

Q146. What does a custom exception class typically inherit from?

- A) `object`
- B) `BaseError`
- C) `Exception`
- D) `Error`

Q147. What is the base class for all Python exceptions?

- A) `Error`
- B) `BaseException`
- C) `Exception`
- D) `PythonError`

Q148. How do you silently ignore an exception and continue?

- A) `skip exception`
- B) `pass` inside `except` block
- C) `ignore exception`
- D) `continue exception`

Q149. What exception does `1/0` raise?

- A) `MathError`
- B) `ZeroDivisionError`
- C) `ArithmeticWarning`
- D) `OverflowError`

Q150. What does `try-except-else-finally` ordering guarantee?

- A) `else` always runs
- B) `finally` always runs
- C) `except` runs before `else`
- D) `else` runs before `except`

SECTION 8 — Object-Oriented Programming

Q151. What keyword is used to define a class?

- A) define
- B) struct
- C) class
- D) object

Q152. What is `__init__` in a Python class?

- A) A destructor
- B) The constructor called when object is created
- C) A static method
- D) A class variable

Q153. What does `self` represent in a class method?

- A) A global variable
- B) The class itself
- C) The current instance
- D) A module reference

Q154. What is inheritance in Python?

- A) Copying code from one class to another
- B) A class deriving properties from another class
- C) Importing a class from a module
- D) Creating multiple objects from a class

Q155. What is encapsulation?

- A) Importing classes from modules
- B) Bundling data and methods within a class restricting direct access
- C) Inheriting from multiple classes
- D) Overriding parent class methods

Q156. What does a name prefixed with `__` (double underscore) indicate?

- A) A public attribute
- B) A protected attribute
- C) A private attribute (name mangling)
- D) A class method

Q157. What is polymorphism?

- A) Inheriting from multiple classes
- B) Defining the same method with different behaviour in subclasses
- C) Creating private attributes
- D) Importing multiple modules

Q158. What does `@classmethod` indicate?

- A) Method cannot access instance variables

- B) Method receives the class (cls) as first argument
- C) A private method
- D) An abstract method

Q159. What does `@staticmethod` indicate?

- A) A method bound to an instance
- B) A method that doesn't receive self or cls
- C) A method that cannot be overridden
- D) A class-level variable

Q160. What does `super()` do?

- A) Makes a method superior
- B) Calls a method from the parent class
- C) Creates a super-user object
- D) Returns the parent class reference

Q161. What is `__str__` used for?

- A) Defining string variables
- B) Returning a human-readable string representation of an object
- C) Converting object to bytes
- D) Initialising string attributes

Q162. What is multiple inheritance?

- A) Creating many objects from one class
- B) A class inheriting from more than one parent class
- C) Inheriting private attributes
- D) Overriding multiple methods

Q163. Which module provides `ABC` for abstract classes?

- A) `abstract`
- B) `abc`
- C) `base`
- D) `interface`

Q164. What is method overriding?

- A) Calling a method twice
- B) Redefining a parent class method in a subclass
- C) Adding parameters to a method
- D) Deleting a method

Q165. What does `@property` do?

- A) Makes an attribute read-only
- B) Allows a method to be accessed like an attribute
- C) Declares a class variable
- D) Creates a static property

Q166. What is composition in OOP?

- A) Inheriting from multiple classes
- B) Building complex objects from simpler objects

- C) Decomposing a class into functions
- D) Combining modules into packages

Q167. What does `__del__` do?

- A) Deletes a class
- B) Is called when an object is garbage collected
- C) Removes an attribute
- D) Destroys a module

Q168. What does `isinstance(obj, ClassName)` check?

- A) If two objects are equal
- B) If `obj` is an instance of `ClassName`
- C) If a class has an attribute
- D) If an object is a primitive type

Q169. What is an abstract class?

- A) A class with no attributes
- B) A class that cannot be instantiated and has abstract methods
- C) A class with only static methods
- D) A class with private methods only

Q170. What does `__repr__` return?

- A) A readable string for end users
- B) An official string representation for debugging
- C) The class name only
- D) The memory address

SECTION 9 — Standard Library

Q171. What does `datetime.datetime.now()` return?

- A) Current date only
- B) Current date and time
- C) Current time only
- D) Unix timestamp

Q172. What module handles high-precision arithmetic?

- A) `math`
- B) `decimal`
- C) `numbers`
- D) `fractions`

Q173. What does `collections.Counter("hello")` return?

- A) 5
- B) A Counter with character frequencies
- C) `['h','e','l','l','o']`
- D) A sorted list

Q174. What does `os.listdir(".")` return?

- A) Current directory path
- B) A list of files and folders in current directory
- C) OS name
- D) Environment variables

Q175. What does `time.sleep(2)` do?

- A) Pauses for 2 milliseconds
- B) Pauses for 2 seconds
- C) Runs code after 2 seconds
- D) Sets a 2-second timeout

Q176. What is the `logging` module used for?

- A) Logging math operations
- B) Recording events during program execution
- C) Storing login info
- D) Measuring performance

Q177. What does `math.floor(3.9)` return?

- A) 4
- B) 3
- C) 3.9
- D) 3.0

Q178. What does `math.ceil(3.1)` return?

- A) 3

- B) 4
- C) 3.1
- D) 3.0

Q179. What does `random.choice([1,2,3,4])` return?

- A) All elements randomly sorted
- B) A randomly selected element from the list
- C) A random int between 1 and 4
- D) A random subset

Q180. What does `os.makedirs("a/b/c")` do?

- A) Creates only directory 'a'
- B) Creates the full directory path a/b/c
- C) Lists directories a b c
- D) Removes directory a/b/c

Q181. What does `pathlib.Path("file.txt").suffix` return?

- A) "file"
- B) ".txt"
- C) "file.txt"
- D) "txt"

Q182. What does `itertools.chain([1,2],[3,4])` do?

- A) Zips the two lists
- B) Links both lists into one iterable
- C) Sorts both lists
- D) Multiplies the lists

Q183. What does `random.shuffle(lst)` do?

- A) Returns a new shuffled list
- B) Shuffles lst in-place
- C) Shuffles and sorts lst
- D) Picks a random element

Q184. What does `math.sqrt(16)` return?

- A) 2
- B) 4.0
- C) 4
- D) 8

Q185. What does `os.path.join("dir", "file.txt")` return?

- A) "dir/file.txt" (platform-specific separator)
- B) "dirfile.txt"
- C) "dir + file.txt"
- D) Error

SECTION 11 — Streamlit Framework

Q186. What command is used to run a Streamlit application?

- A) `python app.py`
- B) `streamlit run app.py`
- C) `streamlit start app.py`
- D) `run streamlit app.py`

Q187. Which function displays a large title in a Streamlit app?

- A) `st.head()`
- B) `st.title()`
- C) `st.big()`
- D) `st.display()`

Q188. Which decorator caches the result of a data-loading function?

- A) `@st.cache_model`
- B) `@st.cache_resource`
- C) `@st.cache_data`
- D) `@st.store_data`

Q189. What does `st.sidebar` allow you to do?

- A) Create a footer
- B) Add widgets to a left-hand panel
- C) Create a new page
- D) Add a top navigation bar

Q190. Which function creates multiple side-by-side columns?

- A) `st.layout()`
- B) `st.columns()`
- C) `st.grid()`
- D) `st.panel()`

Q191. What does `st.dataframe(df)` display?

- A) Edits a DataFrame
- B) An interactive sortable DataFrame
- C) A new empty DataFrame
- D) A static text table

Q192. Which widget creates a dropdown selector in Streamlit?

- A) `st.dropdown()`
- B) `st.picker()`
- C) `st.selectbox()`
- D) `st.choose()`

Q193. What does `st.slider()` create?

- A) An image carousel

- B) A navigation bar
- C) A numeric range input widget
- D) A progress bar

Q194. Which function allows file uploads in Streamlit?

- A) `st.upload()`
- B) `st.file_uploader()`
- C) `st.import_file()`
- D) `st.load_file()`

Q195. What does `st.success("Done")` display?

- A) A red error banner
- B) A yellow warning
- C) A green success message
- D) A blue info box

Q196. What does `st.set_page_config()` control?

- A) Only the page title
- B) Page title, icon, layout and sidebar state
- C) Only the page layout
- D) Background color

Q197. Which layout value makes the app use the full browser width?

- A) `layout="full"`
- B) `layout="wide"`
- C) `layout="expanded"`
- D) `layout="stretch"`

Q198. What does `st.pyplot(fig)` do?

- A) Creates a new matplotlib figure
- B) Displays a matplotlib figure in the app
- C) Saves a figure to file
- D) Imports matplotlib

Q199. Which decorator is best for caching a large ML model?

- A) `@st.cache_data`
- B) `@st.cache_resource`
- C) `@st.cache_model`
- D) `@st.cache_static`

Q200. What does `st.spinner("Loading...")` do?

- A) Rotates an image
- B) Shows an animated spinner while code runs inside a `with block`
- C) Delays execution by 1 second
- D) Creates a circular progress bar

Q201. What does `st.multiselect()` return?

- A) A single selected value
- B) A list of all selected values

- C) A tuple of selected values
- D) A comma-separated string

Q202. How do you display a bar chart directly from a Series or DataFrame?

- A) `st.chart(df)`
- B) `st.bar_chart(df)`
- C) `st.plot_bar(df)`
- D) `df.bar()`

Q203. What does `st.radio()` create?

- A) A volume control
- B) A dropdown menu
- C) A group of mutually exclusive radio options
- D) A checkbox group

Q204. What is `st.session_state` used for?

- A) Saving the app to disk
- B) Persisting variable values across reruns of the app
- C) Storing the source code
- D) Managing server sessions

Q205. What does `st.tabs(["Tab1","Tab2"])` create?

- A) HTML tab characters
- B) Multiple switchable content panes
- C) A navigation sidebar
- D) Table columns

Q206. What does `st.metric("Accuracy", "95%")` display?

- A) A large styled metric card
- B) A progress bar at 95%
- C) A simple text label
- D) A bar chart

Q207. What does `st.write()` do?

- A) Writes content to a file
- B) Displays text, data, or objects intelligently
- C) Prints to the console
- D) Logs a message

Q208. Which function shows expandable/collapsible hidden content?

- A) `st.collapse()`
- B) `st.hide()`
- C) `st.expander()`
- D) `st.toggle()`

Q209. What does `st.progress(50)` display?

- A) Text showing '50%'
- B) A filled progress bar at 50%
- C) A pie chart

D) A loading spinner

Q210. Which of the following displays a STATIC (non-interactive) table?

- A) `st.dataframe()`
- B) `st.table()`
- C) `st.static_table()`
- D) `st.display_table()`

Q211. What default port does Streamlit use?

- A) 3000
- B) 5000
- C) 8501
- D) 8080

Q212. Which platform provides free Streamlit app deployment?

- A) Heroku
- B) Streamlit Community Cloud (share.streamlit.io)
- C) AWS Elastic Beanstalk
- D) Google App Engine

Q213. What does `st.checkbox("Show")` return?

- A) A string 'checked' or 'unchecked'
- B) True if checked False if unchecked
- C) A checkbox widget object
- D) 1 or 0

Q214. What file must be present in a GitHub repo for Streamlit Cloud to install dependencies?

- A) `dependencies.txt`
- B) `packages.txt`
- C) `requirements.txt`
- D) `install.txt`

Q215. What does `st.error("Oops")` display?

- A) A green success box
- B) A yellow warning
- C) A blue info box
- D) A red error message

SECTION 10 — String Operations

Q216. What does "hello world".split() return?

- A) "hello","world"
- B) ["hello","world"]
- C) ("hello","world")
- D) {"hello","world"}

Q217. What does " hello ".strip() return?

- A) " hello "
- B) "hello"
- C) "hello "
- D) " hello"

Q218. What does "hello".replace("l", "r") return?

- A) "herlo"
- B) "herro"
- C) "hrlllo"
- D) "hello"

Q219. What does ", ".join(["a","b","c"]) return?

- A) "a b c"
- B) "a,b,c"
- C) ["a","b","c"]
- D) "abc"

Q220. What does "hello world".find("world") return?

- A) True
- B) 6
- C) "world"
- D) 5

Q221. What does "Python".startswith("Py") return?

- A) False
- B) True
- C) "Py"
- D) 0

Q222. What does "hello".count("l") return?

- A) 1
- B) 2
- C) 3
- D) 5

Q223. What does "HELLO".lower() return?

- A) "HELLO"

- B) "Hello"
- C) "hello"
- D) "hELLO"

Q224. What does "hello"[-1] return?

- A) "h"
- B) "e"
- C) "o"
- D) Error

Q225. What does "abcdef"[1:4] return?

- A) "abc"
- B) "bcd"
- C) "bcde"
- D) "abcd"

Q226. What does "hello world".title() return?

- A) "HELLO WORLD"
- B) "hello world"
- C) "Hello World"
- D) "Hello world"

Q227. What does str.isdigit() return?

- A) The number of digits
- B) True if all characters are digits
- C) True if any character is a digit
- D) A list of digits

Q228. What does "hello" in "hello world" return?

- A) False
- B) True
- C) "hello"
- D) 0

Q229. What does "abc".zfill(5) return?

- A) "abc00"
- B) "00abc"
- C) "abc "
- D) " abc"

Q230. What does " hello ".lstrip() return?

- A) "hello "
- B) "hello"
- C) " hello"
- D) " hello "

ANSWER KEY

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	2	C	3	B	4	C	5	B	6	C	7	C	8	C	9	B	10	C
11	B	12	B	13	C	14	B	15	B	16	B	17	B	18	C	19	B	20	A
21	C	22	B	23	C	24	C	25	B	26	B	27	C	28	C	29	C	30	C
31	C	32	B	33	C	34	B	35	B	36	B	37	C	38	B	39	B	40	C
41	B	42	C	43	B	44	B	45	B	46	A	47	B	48	D	49	B	50	A
51	B	52	A	53	B	54	C	55	B	56	C	57	B	58	B	59	A	60	B
61	B	62	B	63	A	64	B	65	B	66	B	67	B	68	B	69	D	70	B
71	C	72	B	73	C	74	A	75	B	76	C	77	A	78	B	79	B	80	B
81	B	82	B	83	B	84	B	85	B	86	B	87	B	88	B	89	B	90	B
91	C	92	B	93	C	94	B	95	A	96	B	97	B	98	B	99	B	100	B
101	B	102	B	103	A	104	B	105	B	106	B	107	B	108	B	109	B	110	B
111	B	112	C	113	B	114	B	115	C	116	A	117	B	118	B	119	B	120	C
121	B	122	C	123	C	124	C	125	B	126	B	127	B	128	B	129	B	130	B
131	B	132	B	133	B	134	B	135	C	136	B	137	C	138	B	139	B	140	C
141	B	142	B	143	C	144	B	145	B	146	C	147	B	148	B	149	B	150	B
151	C	152	B	153	C	154	B	155	B	156	C	157	B	158	B	159	B	160	B
161	B	162	B	163	B	164	B	165	B	166	B	167	B	168	B	169	B	170	B
171	B	172	B	173	B	174	B	175	B	176	B	177	B	178	B	179	B	180	B
181	B	182	B	183	B	184	B	185	A	186	B	187	B	188	C	189	B	190	B
191	B	192	C	193	C	194	B	195	C	196	B	197	B	198	B	199	B	200	B
201	B	202	B	203	C	204	B	205	B	206	A	207	B	208	C	209	B	210	B
211	C	212	B	213	B	214	C	215	D	216	B	217	B	218	B	219	B	220	B
221	B	222	B	223	C	224	C	225	B	226	C	227	B	228	B	229	B	230	A